Characterizing Memory-Bound Performance in 100 GbE Networks: A Systematic Study of System-Level Optimizations

BRAYTON LORDIANTO, University of California, Berkeley, USA
KAAN KARAKAS, New York University, USA
DIEGO TROY LOPEZ, New York University, USA
ROBERT PAHLE, New York University, USA

We systematically characterize the performance limits of 100 Gigabit Ethernet (100 GbE) networks by evaluating 34 hardware/software configurations combining NUMA affinity, zero-copy mechanisms, MTU settings, congestion control algorithms, and NIC types. Our analysis reveals fundamental shifts in performance bottlenecks at high network speeds, with memory bandwidth—not CPU cycles—emerging as the dominant constraint. We identify three distinct performance regimes: 23–31 Gbps (CPU-bound), 42–50 Gbps (partially optimized), and 57–65 Gbps (memory-optimized). Statistical analysis demonstrates that memory management optimizations dominate CPU scheduling effects, with zero-copy mechanisms achieving extraordinary effect sizes (Cohen's d = 5.23) compared to NUMA affinity (Cohen's d = 1.35). We establish that today's systems achieve no more than 65% of theoretical 100 GbE capacity, identifying a 35 Gbps gap suggesting practical architectural limits for current kernel-based stacks on contemporary hardware. These findings highlight that traditional CPU-focused tuning strategies are inadequate at these speeds and that memory subsystems will increasingly determine throughput in future 200 GbE and 400 GbE networks. Our results provide both optimization guidance for current deployments and architectural insight for future designs.

1 INTRODUCTION

The rapid expansion of data-intensive applications across cloud computing, artificial intelligence, and high-performance computing (HPC) has driven unprecedented demand for high-speed network infrastructure. 100 Gigabit Ethernet (100 GbE) has emerged as a critical technology for modern data centers and HPC systems, providing the high-bandwidth connectivity required for these workloads [5, 15]. However, despite widespread adoption, achieving the theoretical performance of 100 GbE networks remains challenging; real-world deployments often fall significantly short of line-rate throughput due to architectural constraints and configuration inefficiencies.

Traditional network performance optimization has historically focused on CPU processing capabilities as the dominant bottleneck. Yet, recent work highlights a paradigm shift: memory bandwidth and memory hierarchy behavior have overtaken CPU performance as the primary limiting factors in high-speed networking [19, 20]. This is especially seen in AI and machine learning applications where model sizes grow exponentially while accelerator and host memory bandwidth scale sublinearly [3, 9]. Similarly, advances in memory-centric computation reveal new

Authors' addresses: Brayton Lordianto, University of California, Berkeley, Electrical Engineering and Computer Sciences, Berkeley, USA, brayton.lordianto@berkeley.edu; Kaan Karakas, New York University, Courant Institute of Mathematical Sciences, New York, USA, kk4291@nyu.edu; Diego Troy Lopez, New York University, Research Technology Services, New York, USA, diego.lopez@nyu.edu; Robert Pahle, New York University, Research Technology Services, New York, USA, pahle@nyu.edu.

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architectural efforts to mitigate memory bottlenecks in data-parallel workloads [2]. This transition reflects broader architectural trends where CPU capabilities now far exceed system memory access speeds. These findings reinforce the trend first noted in early work where CPU speeds exceeded DRAM capabilities, which is a gap that has only widened over time [6].

The presence of Non-Uniform Memory Access (NUMA) architectures introduces further complexity. NUMA effects can determine whether applications scale efficiently or fail to utilize available hardware resources [18]. NUMA-aware optimizations, including careful memory placement and interrupt routing, have been shown to significantly impact throughput and latency in networking workloads [1, 14]. In parallel, zero-copy and kernel bypass mechanisms have emerged as key optimizations to eliminate expensive memory operations, enabling efficient data transfers that minimize CPU and memory overhead [12].

Despite these advances, current research remains fragmented:

- Many studies explore individual optimization strategies (e.g., NUMA locality, jumbo frames, congestion control algorithms) in isolation rather than systematically examining their interactions [13].
- Most recent breakthroughs, such as NIC DRAM caching or RDMA offload techniques, target applications that can adopt advanced APIs or bypass the kernel entirely [16, 22].
- Comprehensive performance characterizations of traditional kernel TCP/IP stacks in 100 GbE environments, particularly under combined configurations and realistic deployment scenarios, remain scarce.

This paper addresses these gaps through a systematic empirical characterization of 100 GbE performance across a comprehensive space of hardware and software parameters using widely deployed, kernel-based networking stacks. While bypass frameworks like RDMA offer alternative paths, many research and HPC environments remain dependent on kernel TCP/IP stacks for compatibility and ease of deployment.

To conduct this study, we adopt a systematic empirical methodology: we configure and benchmark the throughput from a pair of directly connected 100 GbE servers under a variety of hardware and software settings. We specifically investigate the effects of NIC hardware variations, MTU settings/sizes, interrupt coalescence, NUMA, and TCP congestion control algorithms.The results provide a quantitative characterization of practical performance limits under real-world kernel-based TCP/IP configurations.

This work makes four key contributions:

- **Performance Regime Characterization**: We identify and quantify three distinct throughput regimes (23–31 Gbps, 42–50 Gbps, 57–65 Gbps), showing that they reflect discrete architectural boundaries rather than gradual scaling.
- **Memory Bottleneck Quantification**: We provide statistical evidence that memory management overhead dominates 100 GbE performance limits, with zero-copy optimizations producing extraordinary effect sizes (Cohen's $d > 5.0$) compared to NUMA affinity effects (Cohen's $d \approx 1.4$).
- **Empirical Performance Baselines for 100 GbE Systems**: We systematically evaluate the throughput achievable on modern 100 GbE networks across 34 distinct hardware/software configurations, providing quantitative performance baselines for practitioners working with kernel-based TCP/IP stacks.
- **Architectural Insights for Future Networks**: We show that traditional CPU-focused optimization assumptions break down at 100 GbE speeds and argue that memory subsystem design will become even more critical for 200 GbE and 400 GbE networks.

Our paper provides the first systematic evaluation of combined hardware/software parameter interactions in kernel-based 100 GbE environments, delivering empirical performance baselines and actionable guidance for practitioners. The remainder of this paper is organized as follows: Section 2 reviews related work in 100 GbE optimization and NUMA/memory-aware networking. Section 3 describes our experimental setup. Section 4 presents our results and statistical analysis. Section 5 discusses broader implications and concludes with suggestions for future work.

2 RELATED RESEARCH

As Ethernet speeds scale to 100 Gbps and beyond, optimizing network performance becomes increasingly constrained by architectural bottlenecks rather than raw CPU compute. While prior work explores optimization techniques such as NUMA awareness, zero-copy mechanisms, jumbo frames, and NIC offloading, most studies address these knobs in isolation, often outside the context of kernel-based TCP/IP stacks. This section synthesizes recent work across these domains and identifies gaps that motivate our comprehensive, combinatorial evaluation.

2.1 Memory Bandwidth Bottlenecks and Architectural Limits

Recent studies demonstrate that memory bandwidth—not CPU cycles—has become the primary performance limiter in high-speed networking. Cai et al. [4] show that on 100Gbps links, host stack overhead shifts from protocol logic to data copying, with typical per-core throughput plateauing around 42Gbps on commodity NICs. Additional overheads such as IOMMU and IOTLB pressure has been shown to degrade throughput in 200Gbps networks, though huge-page-backed buffers have been shown to mitigate some of these losses by reducing address translation overheads [8].

In kernel-based TCP/IP stacks, these results highlight a paradigm shift: at line-rate speeds, bottlenecks often arise in the memory hierarchy (e.g., data-copy and buffer management) rather than CPU compute. However, these studies do not quantify how such memory-system limits interact with key optimizations such as NUMA pinning, MTU tuning, or interrupt configuration in 100GbE environments.

2.2 NUMA-Aware Network Optimization

Non-Uniform Memory Access (NUMA) effects significantly impact network performance, particularly for high-bandwidth workloads. Classic work showed how NUMA locality can determine whether applications scale efficiently or experience severe performance degradation in multicore architectures [18].

Recent empirical work further quantifies NUMA's impact in kernel-based network stacks. For instance, in the NESUS (2015) environment, allocating NIC data structures (e.g., Rx/Tx rings) on the same NUMA node as the NIC hardware led to up to 2× throughput improvements, compared to misaligned configurations where NIC resources and memory resided on different nodes [10]. Additionally, it was observed that even a single long-lived TCP flow experiences a 20% drop in throughput per core when the application thread is scheduled on a CPU located on a NUMA node remote from the NIC—due to elevated L3 cache misses and increased data movement overheads [4].

While prior studies offer insight into individual NUMA impacts, they often focus on custom packet-processing frameworks or storage-oriented scenarios.

2.3 Zero-Copy and Memory Avoidance Mechanisms

Zero-copy networking techniques eliminate expensive CPU-based memory copies, enabling data movement directly from application buffers or kernel pages into NIC DMA regions. At high link speeds, especially around 100Gbps, such techniques significantly reduce memory-system bottlenecks. Real evaluations in 100Gbps environments show that enabling Linux-specific APIs like

MSG_ZEROCOPY or using sendfile() can significantly reduce CPU usage and increase throughput, especially compared to traditional read/write loops. For instance, the ESnet INDIS 2024 report demonstrates improved end-to-end transfer rates and efficiency of 30-38% gains using these mechanisms across Intel and AMD platforms, particularly on Linux 5.x and 6.x kernels [21]. Other systems that optimize zero-copy behavior at scale, such as MAIO, can reduce TLB shootdowns and avoid unnecessary remaps, but these are advanced kernel modifications not representative of standard kernel behavior [17].

While previous work highlight significant gains from individual mechanisms, most focus on either API-level strategies or DTN-centered configurations rather than how zero-copy interacts with other tuning strategies—like NUMA affinity, MTU settings, interrupt coalescence, and congestion control—under default kernel TCP/IP stacks in 100GbE benchmarking.

2.4 MTU Optimization and NIC Offload Features

Frame size tuning via jumbo frames (e.g. MTU = 9000 bytes) is a long-standing technique to reduce per-packet processing overhead. Larger frames mean fewer packets for the same data volume, which reduces per-packet CPU overhead and can boost TCP throughput—especially in point-to-point high-bandwidth paths. This efficiency gain is well documented in network tuning manuals (e.g. EPOC/NSF-supported studies), which show that jumbo frames improve throughput and processing efficiency on 100GbE data transfer nodes [7].

Modern NICs further reduce software overhead using offload features such as Large Send Offload (LSO / TSO) and Large Receive Offload (LRO / GRO). These offloads shift segment splitting (LSO/TSO) or packet aggregation (LRO/GRO) into the NIC, reducing CPU demand. For example, InfiniBand-over-Ethernet tests demonstrated up to 45% bandwidth improvement when both LSO and LRO were enabled [11].

That said, actual benefits from jumbo frames and offload features depend on cross-layer interactions: while jumbo frames reduce packet count, the "diminishing performance returns" begin once memory-copy or data-copy overhead becomes the dominant bottleneck—not packet rate alone. This is particularly evident when using kernel-based TCP stacks at 100GbE speeds. Despite operational awareness of MTU and offload tuning, few empirical studies systematically explore their interaction with memory-bound effects.

2.5 Synthesis and Positioning

Many operational and benchmark studies address NUMA affinity, jumbo frames, interrupt coalescence, or zero-copy mechanisms in isolation. However, to date, no published study provides a comprehensive, systematic evaluation of how major host-level optimizations interact in kernel-based 100 GbE networks. Existing work either focuses on microbenchmarks, user-space stacks, or specialized scenarios. Through our exploration, our findings provide essential baselines for practitioners who deploy 100 GbE networks using conventional stacks and offer guidance for architectural decisions in future 200–400 GbE systems.

3 EXPERIMENTAL SETUP

To investigate the performance characteristics of 100 GbE networking across a range of system-level configurations, we conduct a controlled microbenchmark study using two directly connected, identical servers.

3.1 Hardware Configuration

Our experiments consist of 2 Dell C4140 40-core CPU servers, denoted **Node A** and **Node B**. Each server is equipped with:

- Dual-socket Intel Xeon Gold 6230 CPUs (40 cores total, 2.1 GHz base clock),
- 384 GB DDR4-2933 RAM,
- PCIe Gen4 slots with interchangeable NICs.

The nodes are directly connected via 100GbE using QSFP100G-IR4 transceivers over single-mode fiber. No intermediate switching or routing is involved. This topology eliminates external variability and isolates end-host effects. **Node A** functions as the client, while **Node B** functions as the server. All CPU governors are set to **performance** mode across both nodes. All NIC hardware offloading techniques are enabled when available.

3.2 NIC Configurations

We evaluate three 100 GbE network interface cards (NICs) to assess hardware-level variability:

- Mellanox ConnectX-5 EN Dual-Port 100GbE,
- Mellanox ConnectX-6 EN Dual-Port 100GbE,
- Intel Ethernet Controller E810-CQDA2 Dual-Port 100GbE.

All NICs are configured with the latest vendor-recommended firmware and drivers. Hardware offloading features (e.g., LSO, LRO, checksum offload) are enabled by default unless otherwise specified.

3.3 Software and OS Configuration

Both nodes run Ubuntu Server 22.04 LTS with Linux Kernel 5.15.0. The `iperf3` utility (version 3.13) is used to generate transmitted TCP traffic and to measure throughput. TCP timestamps are disabled to reduce clock skew effects. CPU frequency governors are set to `performance` mode, and turbo boost is disabled for stability.

3.4 Measurement Methodology

Each trial consists of a 60-second sustained iperf3 transfer from Node A to Node B. Each configuration is run **5 times**, and we report the average throughput and standard deviation/variance. Unless otherwise noted, NUMA memory binding is applied to the server process to isolate memory locality effects. Interrupt handling is adjusted using `ethtool` and `irqbalance` is disabled when necessary.

3.5 Tested Parameters

The following experimental parameters were varied across the 34 configurations:

- **Maximum Transmission Unit (MTU).** MTU values tested include 1500 bytes (standard Ethernet) and 9000 bytes (jumbo frames). This tests frame processing overhead.
- **Interrupt Coalescence.** Coalescence settings include adaptive (default), 1000 μs, and 1500 μs static delay. This tests trade-offs between latency and CPU interrupt load.
- **NUMA Affinity.** We test default IRQ balancing versus explicitly pinning the iPerf3 server to the same NUMA node as the NIC to enforce local memory access.
- **TCP Congestion Control Algorithms.** We test BBR (default on modern Linux), Cubic, and Reno algorithms to evaluate their impact under high-speed conditions. These are selected as they represent distinct approaches to congestion feedback and are widely used in modern deployments.
- **Zero-Copy.** We enable or disable the use of zero-copy socket flags in iPerf3 to isolate memory copy overhead effects.

index	numa	card_type	zero_copy	window_size	congestion_control	mss	mtu	Mean (Gbps)
0	False	cx6	False	512	bbr	8960	1460	23.015454
1	False	cx6	False	512	bbr	8960	9000	23.106703
2	False	cx6	False	512	cubic	8960	1460	24.664417
3	False	cx6	False	512	cubic	8960	9000	24.955405
4	False	cx6	False	512	reno	8960	1460	24.571907
5	False	cx6	False	512	reno	8960	9000	24.610023
6	False	cx6	True	512	bbr	8960	9000	42.401969
7	False	cx6	True	512	cubic	8960	9000	46.865441
8	False	cx6	True	512	reno	8960	9000	46.902719
9	False	intel	False	512	bbr	8960	1460	25.920273
10	False	intel	False	512	bbr	8960	9000	27.179228
11	False	intel	False	512	cubic	8960	1460	30.521055
12	False	intel	False	512	cubic	8960	9000	28.299921
13	False	intel	False	512	reno	8960	1460	29.403508
14	False	intel	False	512	reno	8960	9000	28.052641
15	False	intel	True	512	bbr	8960	9000	47.772669
16	False	intel	True	512	cubic	8960	9000	50.384702
17	False	intel	True	512	reno	8960	9000	50.213748
18	True	cx6	False	512	bbr	8960	9000	26.259486
19	True	cx6	False	512	cubic	8960	9000	27.883755
20	True	cx6	False	512	reno	8960	9000	27.393837
21	True	cx6	True	256	bbr	8960	9000	57.629294
22	True	cx6	True	256	cubic	8960	9000	65.199096
23	True	cx6	True	256	reno	8960	9000	65.000602
24	True	cx6	True	512	bbr	8960	9000	57.965007
25	True	cx6	True	512	cubic	8960	9000	65.313794
26	True	cx6	True	512	reno	8960	9000	64.88857
27	True	intel	False	512	cubic	8960	9000	30.680605
28	True	intel	False	512	reno	8960	9000	30.504086
29	True	intel	True	256	bbr	8960	9000	60.196446
30	True	intel	True	256	cubic	8960	9000	61.601131
31	True	intel	True	256	reno	8960	9000	62.623043
32	True	intel	True	512	bbr	8960	9000	59.095175
33	True	intel	True	512	cubic	8960	9000	61.940541

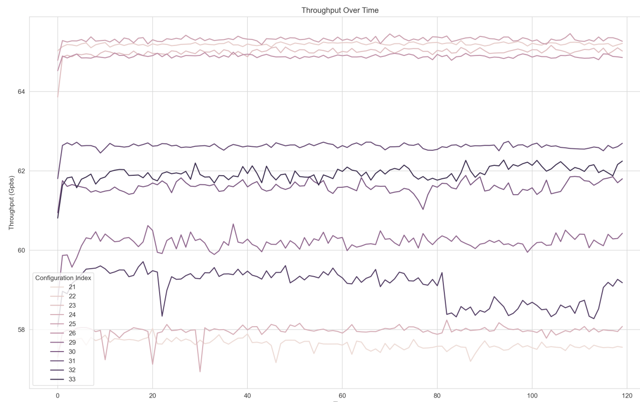
Table 1. Performance results across different configurations

4 RESULTS

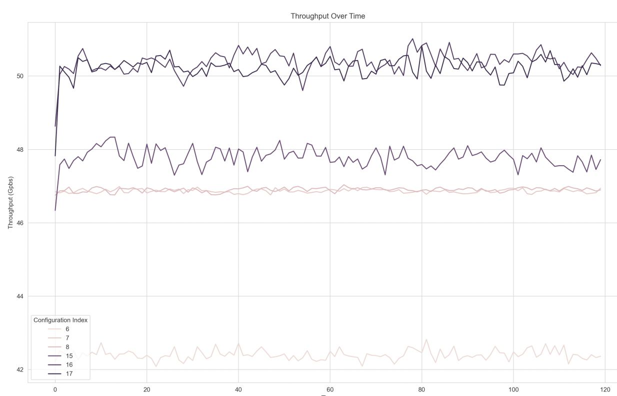
This section presents the throughput characteristics across 34 hardware-software configurations, evaluating the individual and combined impact of NUMA affinity, zero-copy, congestion control algorithm, NIC type, and MTU settings. All configurations are reported in Table 1; summary insights are visualized in Figures 1(a)-3(b).

4.1 Emergent Performance Regimes

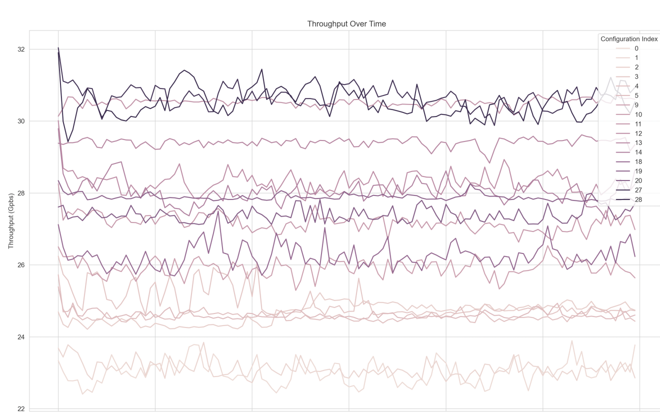
As shown in Figure 1(b), configurations cluster into three distinct throughput ranges:



(a) All Configurations with NUMA and Zero Copy enabled



(b) All Configurations with Zero Copy enabled but NUMA disabled



(c) All Configurations with Zero Copy disabled

Fig. 1. Comparison of different configuration ranges

Feature	Enabled Mean ± SD (Gbps)	Disabled Mean ± SD (Gbps)	Cohen's d
Zero-Copy	56.82 ± 3.66	26.88 ± 1.20	**5.23**
NUMA Affinity	51.51 ± 7.94	33.27 ± 4.90	**1.35**

Table 2. Effect size analysis for key configuration dimensions (Cohen's d). Zero-copy shows an extraordinary impact; NUMA affinity contributes significantly as well.

- **Range C (Low):** 22–32 Gbps, most prominently in Figure 1(c).
- **Range B (Intermediate):** 42– 52 Gbps, though mostly clustering around 48-50 Gpbs, shown in Figure 1(b).
- **Range A (High):** 57–65 Gbps, shown in Figure 1(a).

These are not gradual gradients but distinct performance regimes. Range A configurations consistently exhibit both NUMA pinning and zero-copy enabled (see Fig. 1a), Range B configurations enable zero-copy without NUMA (Fig. 1b), and Range C configurations disable zero-copy entirely (Fig. 1c), regardless of NUMA state.

This suggests that **zero-copy is a threshold-enabling mechanism**, while NUMA primarily refines performance once that threshold is crossed.

4.2 Statistical Analysis Methodology

To quantify the magnitude of performance improvements from key optimizations, we computed Cohen's d effect sizes using the standard formula:

$$d = \frac{\mu_1 - \mu_2}{s_{pooled}} \tag{1}$$

where μ_1 and μ_2 are the sample means for enabled and disabled conditions, and s_{pooled} is the pooled standard deviation:

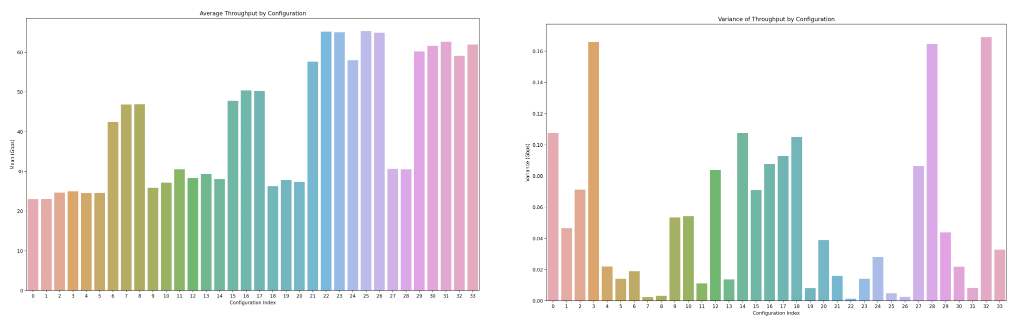
$$s_{pooled} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{2}$$

For zero-copy analysis, we grouped all configurations by zero-copy status (n=17 enabled, n=17 disabled). For NUMA analysis, we grouped by NUMA affinity (n=16 enabled, n=18 disabled). Effect sizes were interpreted using Cohen's conventional thresholds: small (d ≈ 0.2), medium (d ≈ 0.5), large (d ≈ 0.8), and extraordinary (d > 2.0). The results are found in Table 2.

4.3 Effect of Zero-Copy and NUMA Optimization

- **Zero-copy:** Enabling zero-copy yielded a mean throughput of 56.82 Gbps (±3.66), compared to 26.88 Gbps (±1.20) when disabled, across 17 configurations each. This produces a Cohen's $d = 5.23$, classified as an **extraordinary effect size**. It confirms that memory-copy overhead is the dominant throughput limiter in kernel-based TCP/IP at 100 GbE.
- **NUMA:** With NUMA pinning enabled, throughput averaged 51.51 Gbps (±7.94), compared to 33.27 Gbps (±4.90) when disabled (n=16 and n=18, respectively). The resulting Cohen's $d = 1.35$ indicates a **large effect**, though notably less dominant than zero-copy.

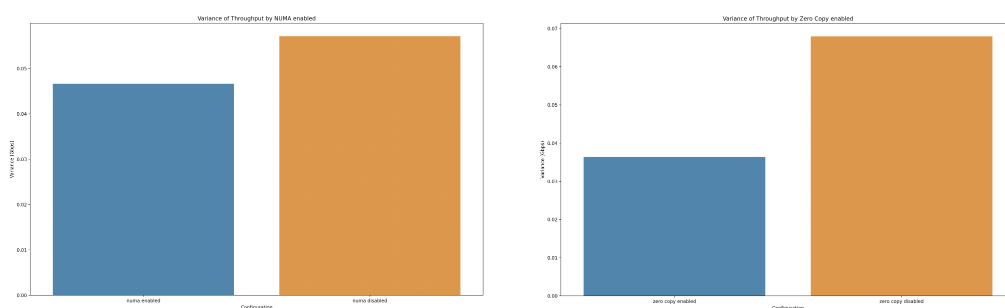
Figure 3 reinforces this: both zero-copy and NUMA optimizations reduce throughput variance, indicating performance stability as well as gain.



(a) Mean Throughput by Configuration Index

(b) Variance Throughput by Configuration Index

Fig. 2. Throughput performance metrics across configurations



(a) Variance Throughput by NUMA enabled

(b) Variance Throughput by zero copy enabled

Fig. 3. Variance analysis by feature enablement

4.4 MTU Size and Frame Processing

MTU settings showed consistent improvements from 1500-byte to 9000-byte jumbo frames, with benefits up to 17% in suboptimal configurations (Range C). However, in zero-copy scenarios (Range A/B), gains from jumbo frames largely diminish. This suggests that **frame-processing overhead becomes secondary once memory copy bottlenecks are eliminated**, a non-obvious interaction not well-characterized in prior literature.

4.5 TCP Congestion Control Algorithms

BBR consistently underperformed Cubic and Reno in high-throughput scenarios. For example, configuration 29 (BBR, zero-copy, NUMA, jumbo frames) achieved 60.1 Gbps, while configuration 25 (Cubic, otherwise identical) reached 65.3 Gbps. We attribute this to BBR's more conservative bandwidth estimation under low-latency, back-to-back connections. Cubic and Reno, in contrast, exploit available bandwidth more aggressively.

4.6 NIC Architecture Effects

Both the Mellanox ConnectX-6 and Intel E810 NICs achieved peak performance, suggesting that driver maturity and hardware offload support are sufficient on both platforms when optimally configured. However, variance in non-optimized scenarios was slightly higher on Intel NICs (see Table I), possibly due to interrupt handling characteristics.

4.7 Summary

While traditional tuning efforts focus on individual knobs, our findings reveal that memory optimization (via zero-copy and NUMA) dominates the performance envelope of 100 GbE kernel TCP/IP stacks. These results underscore a shift in the performance bottleneck from CPU cycles to memory subsystems.

5 DISCUSSION

Our findings suggest a fundamental shift in the performance bottlenecks that shape high-speed networking in kernel-based TCP/IP environments. At 100 GbE speeds, throughput is no longer constrained primarily by CPU availability or protocol overhead but by the system memory hierarchy. This shift reframes traditional optimization assumptions and calls for rethinking both system tuning and network stack design.

5.1 Memory Bandwidth as the Limiting Resource

The most prominent result is the extraordinary effect of zero-copy mechanisms, which achieved a Cohen's $d = 5.23$, a magnitude rarely observed in systems research. This supports a key architectural insight: when line rates exceed 50 Gbps, the cost of memory copying dominates CPU compute cost. Unlike older generations where CPU cycles were the bottleneck, modern multi-core systems increasingly face memory subsystem limits. This is aligned with broader trends in AI, HPC, and RDMA systems where DRAM access lags behind processor and I/O advancements.

5.2 NUMA: Important, but Secondary

NUMA-aware memory placement also yielded significant performance gains (Cohen's $d = 1.35$), but its contribution is contingent upon memory copy behavior. When zero-copy is disabled, NUMA effects are modest and inconsistent. However, when memory copy costs are minimized, NUMA provides a substantial additional benefit by reducing inter-socket memory traffic. This result highlights the importance of co-optimizing both memory bandwidth usage and locality, where isolated tuning of either may produce suboptimal results.

5.3 Performance Regimes and Bottleneck Boundaries

The emergence of three distinct performance regimes (low, middle, high) indicates discrete architectural boundaries, rather than continuous scaling. The lowest regime (23–31 Gbps) was associated with unoptimized configurations; the middle regime (42–50 Gbps) appeared when only zero-copy or NUMA was enabled. Only the high-performance regime (57–65 Gbps) combined both. This segmentation suggests that optimization effects are non-linear and possibly threshold-dependent — below certain thresholds, improvements do not propagate.

Despite aggressive tuning, the maximum observed throughput (65.3 Gbps) still falls 35 Gbps short of 100 GbE line rate. This shortfall reflects deep architectural ceilings, not superficial tuning flaws. It points to a gap that cannot be closed purely through kernel-level TCP optimization.

5.4 Contrast with RDMA and Kernel Bypass

Although this study focused on kernel TCP/IP stacks, it is instructive to contrast our findings with RDMA-based systems. Prior work on RDMA and user-level stacks (e.g., HENS, DPDK) demonstrate near line-rate performance — but require specialized APIs, NIC support, and application modifications. Our findings show that even without kernel bypass, careful tuning can approach 65% of line rate using stock Linux and TCP. This quantifies the performance headroom between traditional stacks and kernel bypass solutions, and helps practitioners understand the tradeoffs involved.

5.5 Implications for 200/400 GbE Networks

As 200G and 400G networks become more prevalent, the architectural bottlenecks we identify will become more acute. Memory systems — not CPUs — will become the dominant constraint. Zero-copy and NUMA-aware designs may not scale linearly without fundamental advances in DRAM throughput or adoption of SmartNIC architectures with local DRAM and offload capabilities. Practitioners designing future networks must plan for these ceilings.

5.6 Limitations and Future Work

Our work is limited to point-to-point configurations using TCP under Linux in controlled settings. The impact of wide-area effects such as RTT, jitter, or packet loss is not considered. We also do not evaluate energy consumption, latency, or system-level cost models. Future work should extend this study to include real-world distributed applications, energy profiling, and comparative evaluations with kernel bypass stacks such as eRPC or RDMA. Additionally, statistical modeling of parameter interactions (e.g., ANOVA or regression) could further explain the contribution of each knob.

Our study also has several important limitations that bound the generalizability of our findings:

- **Network Topology:** Our experiments use a controlled point-to-point topology between identical servers. Real-world deployments involve multiple network hops, heterogeneous hardware, and varying network conditions that may alter the relative importance of the optimizations we evaluate.
- **Traffic Characteristics:** We use synthetic TCP flows generated by iperf3 rather than realistic application workloads. While this enables controlled measurement, application-specific traffic patterns (bursty vs. sustained, bidirectional flows, multiple concurrent streams) may exhibit different sensitivities to NUMA and memory optimizations.
- **Hardware Generalization:** Our results are obtained on specific server configurations (Dell C4140, Intel Xeon Gold 6230). The relative impact of memory vs. CPU bottlenecks may vary on different processor architectures, memory subsystems, or PCIe configurations.
- **Operating System Scope:** All experiments use Linux kernel 5.15 with Ubuntu 22.04. Different kernel versions, TCP stack implementations, or operating systems may exhibit different performance characteristics and optimization sensitivities.

Future work should validate these findings across diverse hardware platforms, realistic application workloads, and multi-hop network environments to establish broader generalizability.

6 CONCLUSION

This paper presents the first comprehensive, empirical analysis of 100 GbE throughput bottlenecks using kernel-based TCP/IP stacks under a systematic parameter sweep. We identify three distinct performance regimes shaped by memory subsystem behavior, with zero-copy and NUMA-aware designs enabling near-maximum achievable throughput (65 Gbps). Our findings reveal that memory bandwidth—not CPU cycles—has become the primary limiter in high-speed networking, a shift with major implications for 200/400 GbE system design. These insights not only guide current optimization practices but underscore the architectural ceiling imposed by memory subsystems—one that future networking research must directly confront.

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